MINI PROJECT – PROJECT REQUIREMENTS DOCUMENT

Course: Java Programming
Project Type: Java Desktop Application using Swing UI & JDBC
Milestone 2: Project Requirements Document & PPT
Marks: 3

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1. PROJECT DETAILS

Project Title: _____

Team Name / Team No.: _____

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2. TEAM DETAILS & HACKERRANK PROGRESS

Sl. No.	Name	Roll No.	Role	HackerRank Profile / Username	Java Stars – Previous Review	Java Stars – Current Review	Progress
1			**Team Lead**				
2			Member				
3			Member				
4			Member				
5			Member				

Note: HackerRank profile details and current Java Stars shall be included in **every project submission and PPT**. Students are expected to demonstrate continuous improvement in Java programming practice throughout the project.

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3. PROBLEM STATEMENT

Clearly describe the problem that the proposed application intends to solve.

The problem statement should explain:

- What is the existing problem?
- Who experiences this problem?
- Why is a software solution required?
- What difficulties exist in the current/manual system?

Problem Statement:

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4. OBJECTIVES OF THE PROJECT

List the main objectives of the proposed application.

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5. PROPOSED SYSTEM

Briefly explain how the proposed Java desktop application will solve the identified problem.

Describe:

- Purpose of the application
- Intended users
- Major operations performed by the system
- How the proposed system improves the existing process

Description:

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6. USERS / ACTORS OF THE SYSTEM

Identify the different types of users who will interact with the application.

User / Actor	Description	Major Activities
Example: Administrator	Manages the application	Add, edit and delete records

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7. FUNCTIONAL REQUIREMENTS

Functional requirements specify **what the system should do**.

Each important functionality shall be listed separately.

Req. ID	Functional Requirement	Priority
FR-01	The system shall allow _____ _____	High / Medium / Low
FR-02	The system shall allow _____ _____	High / Medium / Low
FR-03	The system shall _____ _____ _____	High / Medium / Low
FR-04	The system shall _____ _____ _____	High / Medium / Low
FR-05	The system shall _____ _____ _____	High / Medium / Low
FR-06	The system shall _____ _____ _____	High / Medium / Low
FR-07	The system shall _____	High / Medium / Low

	_____	
FR-08	The system shall _____ _____ _____	High / Medium / Low

Add additional requirements wherever necessary.

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8. MAJOR MODULES / FEATURES

Identify the major functional modules of the application.

Module	Major Functions
Module 1: _____	_____ _____
Module 2: _____	_____ _____
Module 3: _____	_____ _____
Module 4: _____	_____ _____

Example modules may include Login, User Management, Data Entry, Search, Transaction Management, Reports, etc., depending on the project.

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9. INPUT REQUIREMENTS

List the major inputs/data that the application needs to accept from the user.

Input	Description	Validation Required

Students shall identify appropriate validations such as mandatory fields, numeric values, date formats, duplicate entries, valid ranges, etc.

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10. OUTPUT REQUIREMENTS

List the important outputs that the application should generate.

Examples include:

- Search results
- Reports
- Transaction details
- Summary information
- Confirmation/error messages
- Database records displayed through Swing tables

Major Outputs:

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11. DATA / DATABASE REQUIREMENTS

Identify the major information that needs to be stored permanently in the database.

Data / Entity	Important Information to be Stored

Detailed database table design is **not required at this stage**. It will be addressed during the Project Design stage.

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12. NON-FUNCTIONAL REQUIREMENTS

Mention important quality requirements of the system.

Usability

The system shall provide a simple and understandable Swing-based graphical user interface.

Reliability

The application shall correctly store and retrieve information from the database.

Performance

Normal user operations shall be completed without unnecessary delay.

Data Validation

The application shall validate user inputs before processing or storing them.

Error Handling

The system shall display meaningful messages when invalid input, database errors, or other exceptions occur.

Maintainability

The code shall follow proper Java coding standards, meaningful naming conventions and modular design.

Documentation

Important classes and methods shall be properly documented. JavaDocs will be progressively developed during later project reviews.

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13. TECHNOLOGY REQUIREMENTS

Component	Technology
Programming Language	Java
User Interface	Java Swing
Database Connectivity	JDBC
Database	_____
IDE	_____
Version Control, if used	_____

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14. ASSUMPTIONS AND CONSTRAINTS

Mention any important assumptions or limitations of the proposed system.

Assumptions

Constraints

1. The application shall be developed primarily using **Java**.
2. The desktop user interface shall be developed using **Java Swing**.
3. Database operations shall be performed through **JDBC**.

15. EXPECTED FINAL OUTCOME

Briefly describe what the completed project should be capable of demonstrating.

The final application should provide a functional Java desktop solution demonstrating appropriate use of **Java programming, OOP concepts, Swing GUI, JDBC/database operations, exception handling, input validation, coding standards, documentation and testing**.

16. REQUIREMENTS CONFIRMATION

We confirm that the above requirements represent the functionality currently proposed for the mini project. Any major modification to the scope after this milestone shall be discussed during the subsequent project review.

Team Lead: _____

Signature: _____

Date: _____